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Urban Foundation Models: A Survey

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ABSTRACT

Machine learning techniques are now integral to the advancement of intelligent urban services, playing a crucial role in elevating the efficiency, sustainability, and livability of urban environments. The recent emergence of foundation models such as ChatGPT marks a revolutionary shift in the fields of machine learning and artificial intelligence. Their unparalleled capabilities in contextual understanding, problem solving, and adaptability across a wide range of tasks suggest that integrating these models into urban domains could have a transformative impact on the development of smart cities. Despite growing interest in Urban Foundation Models (UFMs), this burgeoning field faces challenges such as a lack of clear definitions and systematic reviews. To this end, this paper first introduces the concept of UFMs and discusses the unique challenges involved in building them. We then propose a data-centric taxonomy that categorizes and clarifies current UFM-related works, based on urban data modalities and types. Furthermore, we explore the application landscape of UFMs, detailing their potential impact in various urban contexts. Relevant papers and open-source resources have been collated and are continuously updated at: <https://github.com/usail-hkust/Awesome-Urban-Foundation-Models>.

CCS CONCEPTS

• **Information systems** → **Spatial-temporal systems**; • **Computing methodologies** → **Neural networks**.

KEYWORDS

Urban foundation models; urban general intelligence; geospatial artificial intelligence; spatio-temporal data mining

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1 INTRODUCTION

Machine learning technologies have become pivotal in transforming urban landscapes, underpinning the development of various smart city services. These technologies enhance urban intelligence by enabling more efficient resource allocation, improved public services, and elevated quality of life for city dwellers [106]. Through analyzing vast datasets from diverse sources such as satellites, urban sensors, and social media platforms, machine learning algorithms identify intricate urban patterns and facilitate accurate forecasting of city dynamics [38, 137]. Such analytical power proves essential in boosting urban operational efficiencies, such as smart energy management, traffic flow optimization, and environmental monitoring, all contributing to smarter and more sustainable cities [56, 89].

The recent advent of foundation models, such as Large Language Models (LLMs, e.g., ChatGPT) and Vision Foundation Models (VFM), has significantly reshaped the research landscape in machine learning and artificial intelligence [151]. These models are characterized by their extensive pre-training on large-scale datasets, which imbues them with unparalleled emergent abilities, including contextual reasoning, complex problem solving, and zero-shot adaptability across diverse tasks [8, 54]. Such abilities make foundation models particularly suitable for interacting with dynamic and multifaceted urban environments, leading towards more integrated, intelligent, and responsive urban systems.

Urban Foundation Models (UFMs), as depicted in Figure 1, represent a novel family of models pre-trained on extensive, diverse urban data sources, encompassing multiple data granularity and modalities. These models exhibit a deep understanding of various urban data types and remarkable adaptability to a wide array of urban tasks [4], significantly contributing towards the ultimate realization of Urban General Intelligence (UGI). By integrating and interpreting diverse urban data types, UFMs can offer comprehensive insights, uncover intricate spatiotemporal patterns, and enhance decision-making across various urban tasks. For instance, UFMs can analyze a combination of human instructions, scene imagery, traffic sensor data, and GPS trajectories to optimize traffic efficiency and safety [20, 57, 136, 149]. Additionally, UFMs can collectively utilize demographic, land use, and environmental data to provide crucial insights for sustainable city development [79, 110, 152].

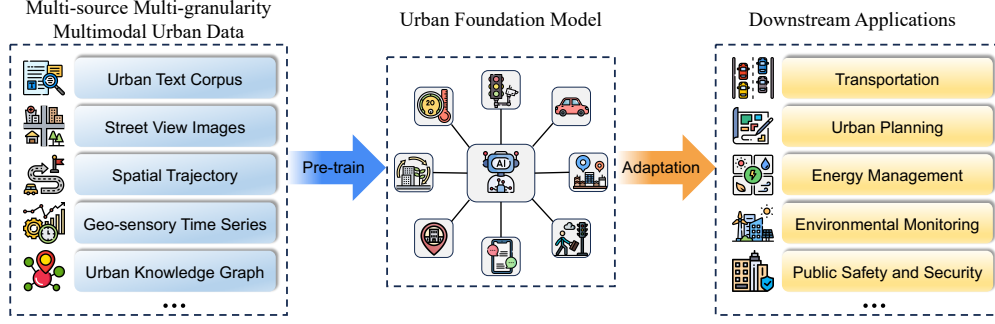


Figure 1: Urban Foundation Models (UFMs) are pre-trained on multi-source, multi-granularity and multimodal urban data and can be adapted for a variety of downstream urban applications.

Despite the burgeoning interest and potential of UFMs, the field faces several challenges, including the absence of clear definitions and the lack of systematic reviews of existing literature [4, 46, 81, 122, 123, 136]. This paper addresses these gaps by presenting a comprehensive survey on UFMs. We start by defining UFMs and discussing their unique challenges. Building upon this, we propose a data-centric taxonomy designed to categorize and encapsulate existing research on UFMs. This taxonomy, rooted in the diverse modalities and types of urban data, aims to highlight the progress and concerted efforts made in this emerging domain. In addition, we also explore the application landscape of UFMs, detailing how they can be applied to enhance different aspects of urban intelligence.

The major contributions of this paper are as follows: (1) We present the first comprehensive and systematic review of Urban Foundation Models, offering a foundational perspective on this evolving field. (2) This work presents the definition of UFMs and delves into the specific challenges associated with their development, shedding light on unexplored aspects. (3) We propose a data-centric taxonomy for UFMs, which categorizes and clarifies existing research based on urban data modalities and types, aiding in the understanding of the field’s current state and future directions.

2 CONCEPTS OF UFMS

UFMs are characterized by their remarkable capacities for understanding, reasoning, and adaptability on multi-source, multi-granularity, and multimodal urban data. We first introduce important basic concepts and then give the formal definition for UFMs.

- **Multi-source.** UFMs are required to integrate large-scale data from diverse urban sources, including sensor networks, satellites, mobile devices, *etc.*, facilitating a comprehensive understanding of urban environments.
- **Multi-granularity.** UFMs are required to handle data at different levels of granularity. At a macro level, they process city-wide patterns like traffic flow and population movement. At a micro level, they need to analyze local trends and behaviors in specific neighborhoods, streets, and individuals.
- **Multimodal.** UFMs are also required to integrate diverse data modalities, such as textual, visual, and sensor-based inputs, enabling them to adapt to various downstream urban applications effectively.

DEFINITION 1 (URBAN FOUNDATION MODELS). *Urban Foundation Models (UFMs) are a family of large-scale models pre-trained on vast amounts of multi-source, multi-granularity, and multimodal urban data. They acquire notable general-purpose capabilities in*

the pre-training phase, exhibiting remarkable emergent abilities and adaptability dedicated to a range of urban application domains, such as transportation, urban planning, energy management, environmental monitoring, and public safety and security.

3 CHALLENGES OF BUILDING UFMS

While UFMs offer significant potential for understanding and managing complex urban environments, building UFMs is a complex task that encounters several unique challenges. These challenges stem primarily from the inherent complexities of urban data, dynamic nature of urban environments, diverse urban task domains, and concerns on privacy and security.

- **Multi-source, multi-granularity, and multi-modal data integration.** The integration of multi-source, multi-granularity, and multi-modal data is one of the primary challenges in building UFMs. UFMs necessitate effectively integrating data from various sources, which provide data at different levels of granularity, from broad city-wide patterns to specific local details. Moreover, the data modalities vary significantly, encompassing text, images, sensor readings, *etc.* Integrating heterogeneous data types presents significant challenges in data preprocessing, normalization, and fusion.
- **Spatio-temporal reasoning capability.** Another major challenge in UFMs is mastering spatio-temporal reasoning. Urban environments are dynamic, with changes and patterns evolving over both space and time. UFMs need to understand and predict complex phenomena that are both time-dependent and spatial-dependent. This involves sophisticated modeling of temporal sequences and spatial distributions, requiring advanced algorithms capable of handling high-dimensional data. This spatio-temporal reasoning skill is particularly vital in urban contexts and is often less emphasized or absent in foundation models from other domains.
- **Versatility to diverse urban task domains.** The versatility of UFMs to diverse urban task domains is another significant challenge. Urban environments are inherently complex and encompass various domains, such as transportation, energy management, and environmental monitoring. While each domain presents unique challenges and requirements, UFMs need to be versatile enough to adapt to these varied contexts. This involves not only tailoring the models to specific domain tasks but also ensuring that they can be effectively generalized across different urban scenarios.
- **Privacy and security concerns.** Deploying UFMs introduces significant privacy and security challenges, stemming from their use of comprehensive urban data from diverse sources, like personal

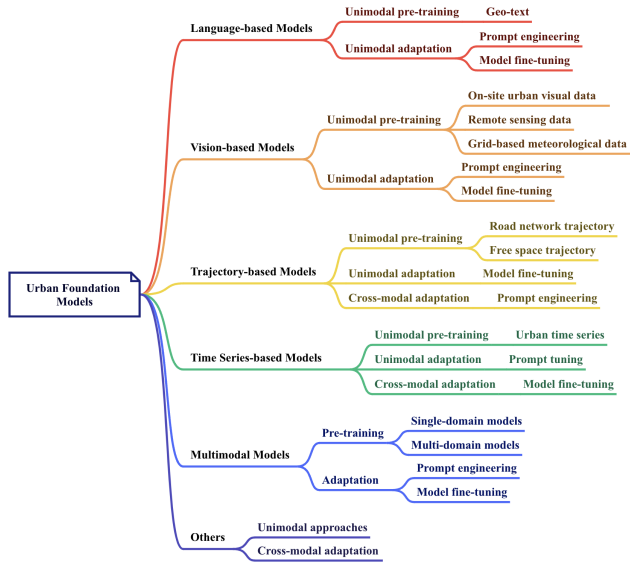


Figure 2: A data-centric taxonomy for existing UFM-related works based on the types of urban data modalities.

devices and social networks. Adherence to privacy laws, such as the General Data Protection Regulation (GDPR) [109], is essential, necessitating strong data anonymization, clear consent for data use, and transparent processes to protect data privacy. Additionally, UFM must ensure data integrity and model security to defense against malicious activities, such as data privacy breaches, data poisoning, jailbreak attacks, adversarial attacks. Given their substantial societal impact, addressing these concerns effectively is crucial for developing UFM that are trustworthy and ethically sound.

4 OVERVIEW OF UFM

As illustrated in Figure 2, we present a data-centric taxonomy for the UFM-related studies based on urban data modalities that UFM process. We will introduce these studies through the lens of their focused pre-training and adaptation techniques.

4.1 Language-based Models

Pre-trained Language Models (PLMs), which typically leverage Transformer models as the backbone and are pre-trained over large-scale unlabeled corpora, have demonstrated strong capabilities in various Natural Language Processing (NLP) tasks [147]. Recently, researchers have found that increasing the parameter scale of PLMs to a significant size (e.g., tens of billions of parameters) not only enhances their capacity but also leads to surprising emergent abilities (e.g., in-context learning, reasoning), which are not present in smaller PLMs (e.g., BERT [21]). These enlarged PLMs (e.g., ChatGPT) are also termed LLMs in literature [151]. Given people living in cities have generated an extensive corpus of textual data, such as documents (e.g., traffic reports), dialogs (e.g., ride-hailing conversations), and geo-texts (e.g., geo-tagged tweets), it is of great value to explore the application of PLMs in the urban context.

4.1.1 Unimodal Pre-training. Recently, a few studies [23, 45] focus on training PLMs by only using domain-specific data, outperforming generalist counterparts on tasks within those domains. For

example, ERNIE-GeoL [45] is a pre-trained language model dedicated to improving geography-related tasks at Baidu Maps. The model is pre-trained on massive textual data extracted from a heterogeneous graph that contains rich toponym and spatial knowledge, by using masked language modeling and geocoding tasks. Ding et al. [23] propose MGeo, a language model specifically pre-trained on abundant geographic context data for query-POI matching [135]. MGeo designs a geographical encoder to encode geographic context representations. It also utilizes masked language modeling and contrastive learning in pre-training stage. However, large-scale pre-training requires substantial computational resources and electricity, which may be difficult for researchers to obtain.

4.1.2 Unimodal Adaptation. Instead of training urban language models from scratch, previous studies mainly focus on adapting existing PLMs to urban scenarios. In this way, we can make full use of the world knowledge encapsulated in PLMs while significantly reducing the demanded computational cost. Specifically, there are two categories of adaptation approaches: prompt engineering and model fine-tuning.

- Prompt engineering. Prompting aims to steer PLM’s behavior toward desired outcomes via pre-defined text inputs, i.e., task description or a set of demonstrations, which allows PLMs to handle tasks they have never explicitly been trained for. Prompting offers a flexible, efficient, and user-friendly way to harness the capabilities of PLMs without the necessity of retraining or fine-tuning the model. Several preliminary studies [5, 34, 47] suggest that PLMs have encoded a wealth of urban factual and spatio-temporal knowledge from their training corpora. These knowledge can be effectively queried by prompting techniques. For instance, Prabin et al. [5] investigate a set of geospatial skills of LLMs, including geospatial knowledge, awareness, and reasoning. By leveraging appropriate prompts, LLMs can utilize the encoded geospatial knowledge to benefit diverse tasks, such as deriving coordinates of cities and understanding urban entity relationships [88].

Building upon the above findings, recent studies have explored the capabilities of LLMs in a variety of urban applications, such as urban science research [28], travel planning [93, 120, 121], and urban question answering [86]. To name a few, Aghzal et al. [2] systematically evaluate the reasoning abilities of LLMs in path planning tasks through different few-shot prompting strategies. GeoLLM [83] focuses on extracting geospatial knowledge from LLMs by using prompts augmented with auxiliary map data, achieving superior performance in real-world geospatial prediction tasks. Mai et al. [80, 81] demonstrate that LLMs can act as zero-shot or few-shot urban task learners via prompting and outperform tailored models in toponym recognition, country-level time series forecasting, and urban function classification. In addition, Zheng et al. [149] explore the potential of LLM prompting in various traffic safety applications, including accident reports automation, traffic data augmentation, and multisensory data analysis.

While direct prompting is flexible and resource-efficient, it demands powerful, generalist LLMs and rich contextual information to achieve satisfactory performance, which may restrict the applicability of LLMs in complex real-world applications.

- Model fine-tuning. Generalist PLMs, pre-trained on vast amounts of Web textual data, might lack sufficient knowledge for specific

tasks or domains. Some studies focus on fine-tuning PLMs with domain-specific data to enhance model performance towards particular areas. SpaBERT [61] extends BERT to capture the spatial varying semantics of geo-entities by fine-tuning synthetic sentences from geographic databases for geo-entity comprehension. It also introduces a spatial coordinate embedding module to preserve spatial relationships of geo-entities in latent feature space. GeoLM [64] proposes to jointly learn linguistic and geospatial context via contrastive learning and masked language modeling. Compared with SpaBERT, GeoLM can effectively retain general linguistic information. Mei et al. [84] introduce a fine-tuned BERT model to enhance the query intent understanding in POI retrieval tasks. The model is initialized from the pre-trained weights of BERT, and fine-tuned based on the knowledge from both the user behavior logs and the POI corpus. More recently, there have been efforts to fine-tune open-source LLMs for urban applications. For example, PlanGPT [153] is proposed to improve the comprehension and reasoning capabilities of LLMs in urban and spatial planning domain. In particular, it utilizes a two-stage fine-tuning approach to inject interdisciplinary urban knowledge into LLM and enhance its ability to adeptly process government documents.

Despite fruitful progress, these efforts mainly focus on pre-training language models or leveraging existing PLMs for specific urban applications (e.g., map services). PLMs specialized for a broad range of urban tasks and domains have not been reported in the literature, which could be a potential future direction.

4.2 Vision-based Models

Existing Vision Foundation Models (VFM) mainly fall into two categories: language-augmented VFMs and vision-only VFMs. Language-augmented models, such as CLIP [91], aim to learn an image encoder by aligning paired text and images with contrastive learning. Conversely, vision-only VFMs are trained on purely visual content data through self-supervised pre-training, such as masked autoencoder (MAE) [41] and LVM [3]. These emerging VFMs, pre-trained on general visual data, exhibit robust generalization and impressive zero-shot learning abilities across a wide range of computer vision tasks. However, since the distributions of urban visual data (e.g., satellite images) are significantly different from general visual data [46], existing VFMs might not be suitable for handling urban-centric tasks. Consequently, building tailored urban VFMs becomes highly necessary and practical.

4.2.1 Unimodal Pre-training. With the widespread deployment of camera and satellite technologies, a vast volume of visual data has been collected in urban spaces. The massive availability of data enables a series of research to train large urban vision models from scratch. We classify existing models based on the type of pre-training data they utilize: on-site urban visual data, remote sensing data, and grid-based meteorological data.

- **On-site urban visual data.** On-site urban visual data, such as street-view imagery [7] and surveillance footage [9], are generated by ground-level devices (e.g., smartphones, cameras, and automotive LiDARs) deployed in cities. Recently, several pre-training methods [59, 74, 118] were proposed to learn representations from unlabeled on-site urban visual data. For instance, Urban2Vec [118] applies the Tobler’s First Law of Geography [85]—which posits

that “everything is related to everything else, but near things are more related than distant things”—to develop a contrastive learning framework. This framework learns representations of street-view images by enforcing that spatially adjacent images are similar in latent feature space. Moreover, KnowCL [74] devises a pre-training framework by maximizing the mutual information between street-view images and their corresponding knowledge graph.

- **Remote sensing data.** Remote Sensing (RS) data are typically captured from satellites, aircraft, or drones. Existing RS foundation models adopt the Masked Image Modeling (MIM) [41] paradigm, which first encodes a masked RS image with an image encoder, and then decodes the unmasked portion to reconstruct the entire image. Following this, Wang et al. [112] train a plain Vision Transformer (ViT) with 100 million parameters to solve various RS tasks, ranging from detecting urban objects to identifying region functions. Notably, a rotated varied-size window attention is proposed to replace the standard attention block in ViT, significantly reducing the computation cost for high-resolution RS images. Cha et al. [10] further examine the impact of scaling up ViT to one billion parameters in the RS domain. Additionally, RingMo [101] leverages Swin Transformer [75], a well-known variant of ViT, as the model backbone, which can effectively capture dense and small objects that are often overlooked in RS scenarios. RingMo-Sense [132] further extends RingMo for RS spatiotemporal prediction tasks by leveraging a multi-branch structure with different masking pre-training strategies to learn multi-scale spatiotemporal representations.

- **Grid-based meteorological data.** Grid-based meteorological data can be naturally represented as images, where pixel intensities characterize the spatial distribution of meteorological variables, e.g., temperature, humidity, and pressure. In recent years, the rapid development of VFMs has been making an important impact on weather and climate research [13]. As a pioneering study, Pangu-Weather [6] proposes a 3D Earth-specific transformer for medium-range weather forecasting, outperforming the world’s best Numerical Weather Prediction (NWP) system on reanalysis data. Subsequently, Nguyen et al. [87] present ClimaX, the first foundation model for climate and weather analysis. It improves the vanilla ViT architecture with decoupled variable encoding and aggregation strategies, which enhance the model’s adaptability for heterogeneous atmosphere variables. ClimaX also generalizes well to scenarios involving regions and variables not encountered during pre-training. Different from the above models trained in an autoregressive manner, W-MAE [82] introduces the concept of MAE into the weather and climate domain. This model is pre-trained by reconstructing the spatial relations among meteorological variables, which can be fine-tuned to benefit downstream forecasting tasks [139].

4.2.2 Unimodal Adaptation. Sometimes there is insufficient urban visual data to pre-train a large-scale VFM from scratch. Thus, some studies directly adapt off-the-shelf VFMs, pre-trained on general visual data, to urban tasks. The adaptation can be achieved through prompt engineering and model fine-tuning techniques.

- **Prompt engineering.** Current research in this category mainly focuses on leveraging the Segment Anything Model (SAM) [55] for urban RS image segmentation. SAM is a foundation model tailored to segmentation tasks, which consists of (1) a ViT-based image encoder for generating image embeddings; (2) a prompt encoder

that processes user prompts (specify what to segment in an image) to create embeddings; (3) a lightweight mask decoder that combines image and prompt embeddings to predict segmentation masks. Particularly, the user prompt can range from sparse (e.g., points, boxes, and texts) to dense (e.g., coarse-grained masks). For example, Wang et al. [111] propose SAMRS for RS segmentation by directly harnessing the zero-shot capabilities of SAM. SAMRS manually designs six basic prompts based on the characteristics of RS images, and leverages the optimal combination of prompts identified in experiments for segmentation. RSPrompter [12] further simplifies the process by automating prompt generation. It generates appropriate prompts, such as point or box embeddings, for SAM input by analyzing the hidden layers of the encoder.

- **Model fine-tuning.** In urban scenarios, model fine-tuning is utilized to improve a pre-trained VFM for a certain capability, e.g., semantic segmentation. For instance, GeoSAM [97] enhances SAM's performance for mobility infrastructure segmentation using a fine-tuning strategy with the help of automatically generated visual prompts. RingMo-SAM [127] adopts fine-tuning to update the prompt encoder of SAM for multi-source RS segmentation. Moreover, StreetCLIP [35] fine-tunes CLIP with synthetic captions via meta-learning for street-view image geolocalization tasks.

So far, existing studies mainly focus on relatively small datasets, and the pre-training of VFMs on large-scale urban visual data has not been well studied yet. On the other hand, the joint learning of urban visual data and other data modalities may enhance the visual comprehensive understanding of urban context, which is also an interesting and beneficial direction for future research.

4.3 Trajectory-based Models

Trajectory data is an important modality type prevalently existing in location-based urban tasks, such as driving behavior analysis [31, 43, 58, 129] and human mobility analysis [27, 48, 66, 67, 70]. Formally, trajectory data is a specific type of geo-sensory data that can be defined as a sequence of time-location points, $T = \{(t_1, p_1), (t_2, p_2), \dots, (t_n, p_n)\}$, where for each point, t_i is the timestamp and p_i is the corresponding location in space, often represented as geographic coordinates.

4.3.1 Unimodal Pre-training. Trajectory data typically requires tailored modeling to suit specific applications [27, 58]. Unimodal pre-training, usually involving self-supervised learning on large pure trajectory datasets, enables models to capture intrinsic features and patterns of trajectory data, enhancing their versatility across different applications. We introduce the pre-training methods on two types of trajectory data: road network trajectories and free space trajectories.

- **Road network trajectory.** Road network trajectory refers to trajectories that are confined within road networks, e.g., the paths of pedestrians or vehicles on streets and highways. Learning low-dimensional embeddings of trajectories and paths is one of the most crucial tasks in road network trajectory pre-training which can benefit various downstream tasks. T2vec [60] and Traj2vec [131] represent early attempts at trajectory-based pre-training models, adopting the sequence-to-sequence [102] encoder-decoder framework to learn trajectory representations. These models are pre-trained

by reconstructing trajectory sequences through a generative self-supervised task. However, these methods process raw trajectories as input, potentially compromising the quality of trajectory representation due to noise. To mitigate this, Trembr [29] projects raw trajectories onto road networks, using road segment sequences as inputs. Furthermore, the decoder in Trembr is tailored to reconstruct both road segments and their associated travel times. To further refine trajectory representations, Toast [16] initially learns road representations from the semantic information of road networks, subsequently generating trajectory representations using a pre-trained bidirectional Transformer encoder. Unlike prior works, LightPath [130] directly learns path representations for diverse downstream tasks, applying a self-supervised relational reasoning method to expedite the training of path encoders.

While the aforementioned methods utilize generative or contrastive approaches in pre-training, MMTEC [66] contends that these methods introduce biases in trajectory embeddings. To mitigate this, MMTEC employs a unique pre-training task using maximum entropy coding to reduce biases in trajectory embeddings. Additionally, some studies combine both generative and contrastive pre-training approaches. HMTRL [70] is a route representation learning framework tailored for multi-modal transportation recommendations. The framework's spatiotemporal pre-training strategy, featuring masked attribute prediction and trajectory contrastive learning, bolsters its generalization capabilities. START [48] also introduces two self-supervised tasks for pre-training a trajectory representation model: span-masked trajectory recovery and trajectory contrastive learning.

- **Free space trajectory.** Free space trajectory refers to trajectories in open, unstructured spaces without the constraints of road networks, such as pedestrian movements in parks or open areas. A representative subset of this category is check-in trajectories, where individuals record their location at specific POIs, often using social media or location-based services [77]. Several approaches have focused on simulating and understanding human mobility in unstructured environments. CTLE [67] presents a pre-training model inspired by BERT's masked language model task, generating location embeddings by considering contextual neighbors within trajectories, effectively addressing the multi-functional properties of locations. CACSR [31] is an adversarial contrastive model that enhances learning of check-in sequence representations through a contrastive learning-based pre-training task for its encoder.

4.3.2 Unimodal Adaptation. Unimodal adaptation for trajectory-based foundation models refers to transferring the pre-trained models between different trajectory datasets. Current works primarily focus on adapting a model trained on trajectories from one city to another, through model fine-tuning. Reformd [32] employs standard transfer learning procedures, transferring model parameters learned on a data-rich region to a target region with scarce data. Axolotl [33] also focuses on cross-region model transfer to enhance POI recommendations in data-scarce regions, and employ cluster-based knowledge transfer without the need for any common users or POIs between regions. Building on this idea, CATUS [98] addresses data scarcity and imbalance in POI recommendations across cities by transferring category-level universal transition knowledge through two self-supervised tasks in the pre-training phase.

4.3.3 Cross-modal Adaptation. This line of research aims to transfer learned patterns, features, or knowledge from other data modalities to trajectory. Recent advancements in LLMs (e.g., GPT-4 [1], Llama2 [105]) have demonstrated exceptional zero-shot learning and generalization capabilities across various domains. In the realm of trajectory analysis, several studies have utilized LLMs for trajectory-based tasks.

A number of research have investigated the use of LLMs in tackling tasks related to human mobility by primarily employing prompt engineering techniques. LLM-Mob [116] designs effective prompts to enhance LLMs' reasoning on mobility data, producing both accurate and interpretable results on human mobility prediction tasks. LLM-MPE [65] seeks to use LLMs for forecasting human mobility in complex scenarios, such as public events, employing a chain-of-thought prompting approach to base predictions on historical mobility patterns and event descriptions. LLMob [113] presents an LLM agent framework for personal mobility generation, involving aligning LLM to urban mobility data by self-consistency approach and interpreting the activity generation via retrieval augmentation.

Current trajectory pre-training mainly uses models with small parameters, under-exploring the emergent abilities [119] of larger models and data sizes like seen in LLMs, indicating opportunities for discovering new abilities in managing complex trajectory data in diverse urban contexts. Additionally, while cross-modal adaptation often involves directly prompting LLMs to process trajectory data, employing fine-tuning techniques such as prompt tuning and model fine-tuning could lead to a more thorough adaptation of LLMs to trajectory-related tasks [125].

4.4 Time Series-based Models

Time series data are ubiquitous due to the temporal characteristic of urban environments, associated with diverse urban tasks such as traffic forecasting [38, 69, 137, 140], air quality forecasting [36, 37, 39], energy consumption prediction [124]. In urban contexts, time series (*a.k.a.* spatio-temporal data in some studies [72, 96, 134]) can be formally denoted as $X \in \mathbb{R}^{T \times N \times C}$, where T, N, C represent the number of time steps, variables, and channels, respectively

4.4.1 Unimodal Pre-training. Unimodal pre-training typically involves training deep neural networks on numerous large time series datasets in a supervised or self-supervised manner. Through pre-training, the model learns to capture the intricate temporal patterns and inter-variable correlations within the data.

Beyond pre-training on ordinary time series that usually only considers the temporal dimension [24–26, 30, 73, 146], the studies on urban time series emphasize on incorporating the spatial dimension to explicitly accommodate the dependencies between variables. Some works [114, 133, 145] model the spatial relations as a grid map with fixed sizes and employ Convolution Neural Networks (CNNs) to encode the spatial pattern. For example, ST-GSP [145] proposes a semantic flow encoder with ResNet as the backbone to capture the spatial dependencies. Besides, it devises a Transformer encoder with an early-fusion paradigm to capture multi-scale temporal information for reconstruction-based pre-training and downstream urban flow prediction. Lately, UniST [134] further proposes a universal foundation model for spatio-temporal prediction that could generalize diverse data formats and variations of data distributions.

It designs a generative pre-training strategy with four distinct masking techniques and a prior knowledge-guided prompt network for adaptation. However, grid-based spatial modeling is limited to local regular dependencies and fails to exploit irregular and arbitrary variable-wise connections.

In light of that, recent works [51, 72, 76, 90, 103] focus on graph-based spatial encoding and usually use Spatio-Temporal Graph Neural Networks (STGNNs) to extract the spatio-temporal information. For instance, due to data scarcity, STGCL [72] devises a contrastive pre-training framework for traffic forecasting. It introduces node- and graph-level contrast paradigms with several data augmentation techniques and debiased negative filtering strategies from both spatio-temporal views. As vanilla STGNNs typically focus on short-term time series, STEP [96] devises TSFormer for time series pre-training to capture long-term temporal patterns with segment-level representations. TransGTR [52] focuses on how to learn the graph structure in a transferable way, which pre-trains a structure generator and transfers both the generator and the spatiotemporal forecasting module across cities. Further, MC-STL [141] spots the spatiotemporal heterogeneity (*i.e.*, spatial correlation and region importance are time-varied), and proposes two pre-training strategies: spatial masked reconstruction and temporal contrastive learning, to understand inter-region correlations and regional pattern variations over time. GPT-ST [63] pre-trains a spatio-temporal mask autoencoder by incorporating a parameter customization scheme and adaptive mask strategy, utilizing a hierarchical hypergraph structure to capture multi-level spatial dependencies from a global perspective.

4.4.2 Unimodal Adaptation. Unimodal adaptation aims to tailor the pre-trained time series foundation models to new time series datasets to enhance downstream task performance. Due to the emergence of prompt tuning methods in data modalities like natural language, vision, and graph [100], prompting methods for time series also receive growing attention. For example, PromptST [143] adds tunable prompts to spatiotemporal data to efficiently fine-tune the pre-trained traffic forecasting model and avoid the catastrophic forgetting issue. MetePFL [14] and FedWing [15] focus on federated weather forecasting and use prompting techniques for personalized federated learning to reduce communication overheads and enhance privacy protection.

4.4.3 Cross-modal Adaptation. Due to insufficient time series data to train a foundation model from scratch, cross-modal adaptation approaches directly adopt the foundation models pre-trained on other modalities and aim to transfer the modality-sharing knowledge to time series tasks. Recently, most cross-modal studies have focused on adapting pre-trained LLMs via partial parameter fine-tuning for time series to sufficiently leverage the remarkable abilities of LLMs [49, 50].

To adapt the LLM for urban time series, GATGPT [17] uses a graph attention module for spatial graph encoding and passes the spatio-temporal embeddings to the pre-trained GPT-2 model for downstream spatio-temporal imputation. TPLLM [92] adopts 1-D convolutional neural networks and graph convolutional networks [78, 138] to enable LLMs to understand spatio-temporal patterns in traffic data. Additionally, STG-LLM [71] proposes a spatio-temporal tokenizer to transform spatio-temporal data into tokens

and an adapter architecture for efficient adaptation, which facilitates the understanding of pre-trained LLMs. Similarly, ST-LLM [68] treats the time steps at each spatial location as tokens, and fuse various spatial and temporal prior information into input embedding to facilitate traffic prediction. To align the spatio-temporal dependencies with the comprehension space of LLMs, UrbanGPT [62] proposes a spatio-temporal instruction tuning paradigm that aligns spatio-temporal and textual information, enhancing multiple spatio-temporal prediction tasks, particularly in zero-shot scenarios.

The existing studies are mostly limited to homogeneous time series data formats, e.g., regularly sampled time series, and specific domains, which may disobey the universal property of foundation models [30]. Building UFM that can be applied to heterogeneous data formats and diverse domain tasks remains an open problem.

4.5 Multimodal Models

Due to the intricate nature of urban environments characterized by diverse data types, there is growing attention on developing multimodal foundation models to handle urban tasks. By integrating these disparate data modalities, multimodal foundation models aim to achieve a more holistic understanding of urban dynamics, largely enhancing various urban applications and setting the stage for urban general intelligence.

4.5.1 Pre-training. To develop a comprehensive understanding of urban environments, multimodal UFM aim to systematically pre-train models on extensive multimodal datasets containing urban-specific information such as textual, visual, and geographic data. These works can be classified into single-domain models and multi-domain models.

- **Single-domain models.** Among the existing pretraining-based studies, some works focus on improving a specific urban domain such as urban profiling and transportation. UrbanCLIP [126] combines LLMs with satellite imagery for urban profiling. It leverages LLMs to generate textual descriptions for satellite images and employs contrastive learning as well as language modeling loss as supervision for urban visual representation learning of the model. TengYun [144] is a transportation foundation model developed for TransVerse, focusing on cognition, operation, management, and control. It combines pre-training with prompt tuning for training, starting with pre-training on cloud servers using the Transformer framework and deep reinforcement learning to process historical multi-modal data. During prompt tuning, edge servers customize prompts for different tasks and strategies by treating training as a self-supervised regression problem.

- **Multi-domain models.** While the above efforts have been made in pre-training multimodal UFM, their focus on training models using data specific to a single urban domain can limit their application to broader urban tasks. Therefore, some works attempt to construct multimodal UFM that can be applied to multiple urban domains. CityFM [4] is a framework that pre-trains foundation models by using spatial, visual, and textual data from OpenStreetMap in a self-supervised manner. It utilizes contrastive objectives to learn multimodal representations of geospatial entities, and demonstrate effectiveness in downstream tasks such as traffic speed inference and building functionality classification. AllSpark [95] introduces

the Language as Reference Framework (LaRF) to develop a multi-modal spatio-temporal model that amalgamates thirteen distinct modalities into a single framework. LaRF's concept centers around correlating abstract concepts from each modality with language, facilitating a collective interpretation within the language-based unified representation space.

4.5.2 Adaptation. Another line of work explores introducing foundation models from other domains, e.g., LLMs, to address urban challenges. The adaptation approaches of these works primarily include prompt engineering and model fine-tuning.

- **Prompt engineering.** The works using prompt engineering usually combine foundation models with domain-specific tools like Geographic Information Systems (GIS) and external databases. TrafficGPT [136] is a framework for urban traffic management that integrates ChatGPT and various external tools. It involves prompt management, task understanding and planning through LLMs, and leverages LLMs to invoke available tools for specific executions (e.g., database retrieval and analysis, data visualization, and system optimization). TrafficGPT also concludes with dialogue memory storage to enhance its conversational context for future interactions. OpenTI [20] uses prompt engineering to enhance language models' understanding and execution abilities and integrate multiple traffic simulation platforms like SUMO and CityFlow. It supports complex simulation tasks such as map data processing from scratch, traffic signal control, and demand optimization. GeoGPT [142] is a framework integrating LLMs with mature GIS tools to assist non-professionals in solving complex geospatial tasks using natural language. It leverages LLMs to understand user demands and conducts automated data collection, processing, and analysis in GIS, enabling users to perform tasks like spatial queries and facility siting. Besides, Zhou et al. [152] apply LLMs to simulate planners and residents for participatory urban planning. This work presents a planning workflow where the planner proposes an initial land-use plan based on the image map and textual description of the region, and this plan is then revised through discussion with residents.

- **Model fine-tuning.** Beyond directly invoking the off-the-shelf foundation models, some studies also introduce a fine-tuning process for better adapting other domain foundation models to urban applications. TransGPT-MM [115] is fine-tuned from the general multi-modal LLM, VisualGLM-6B [22], to cater to the transportation domain specifically. This adaptation involves fine-tuning LLM on multi-modal domain-specific datasets derived from driving tests, traffic signs, and landmarks, showcasing powerful capabilities in traffic-related analysis and modeling tasks. VELMA [94] is an embodied LLM-based agent for vision and language navigation in the street view. It integrates visual observations from street view through CLIP for landmark recognition and uses a verbalization pipeline to translate visual information and navigation instructions into textual prompts for decision-making. It fine-tunes LLM on urban domain training instances using LoRA [42], which adapts attention query and value projections in the LLM. Xu et al. [123] integrates LLMs into urban systems to tackle urban challenges, using continued pre-training to infuse the CityGPT model with urban knowledge. It combines general texts, domain-specific data (e.g., urban knowledge graphs and geographic data), and task-solving processes, and then employs supervised fine-tuning to tailor CityGPT more closely to specific urban tasks and human preferences.

Despite works on multimodal UFM proactively advancing to handle multimodal urban data, they are primarily limited to integrating a restricted range of urban data types or applying to only a narrow scope of urban domains. Furthermore, the potential for spatio-temporal reasoning and the privacy and security issues in UFM remain largely unexplored. These reveal significant opportunities for creating more versatile, practical, and trustworthy UFM.

4.6 Others

Except for the aforementioned modalities, there exist some studies exploring the foundation models in other urban scenarios, such as geographical data [104], road network [11] and control policy [57].

4.6.1 Unimodal Approaches. OpenStreetMap (OSM) is a vital information source of geographic information worldwide [53]. Nicolas et al. [104] provide GeoVectors which is a world-scale corpus of OSM entity embeddings, to enable the convenient usage of geographic information for machine learning algorithms. SARN [11] proposes a contrastive-based graph pre-training method for spatial structure-aware road network embedding, to facilitate downstream road property prediction, trajectory similarity prediction, and shortest-path distance prediction tasks.

4.6.2 Cross-modal Adaptation. There emerge some studies on leveraging the capacities of LLMs for challenging urban tasks, such as traffic control [19, 57, 108]. For example, PromptGAT [19] uses LLMs to generate human knowledge, that helps the neural networks understand realistic cases (e.g., weather conditions) in traffic signal control tasks [99]. LLMLight [57] proposes a pioneering framework that utilizes LLMs as control agents to enable human-like decision-making in traffic signal control tasks, showcasing remarkable effectiveness, interpretability, and generalization ability.

5 APPLICATIONS

In this section, we discuss several urban application scenarios that could significantly benefit from UFM.

- **Transportation.** UFM enable a new generation of intelligent transportation systems, supporting humans throughout traffic management workflow and markedly reducing workloads. A few studies have attempted to develop foundation models for transportation, such as TransGPT [115] and TFM [117]. However, these approaches primarily focus on specific aspects of transportation, e.g., traffic forecasting. Developing a versatile UFM capable of handling multimodal traffic data (e.g., time series, trajectories, images, and text) and diverse transportation tasks still remains an open problem.

- **Urban planning.** The widely available urban data, such as road network structure and human mobility, has indeed uncovered the underlying issues of a city, providing urban planners with valuable insights for making informed decisions [150]. UFM can bring substantial benefits to urban planning. On the one hand, they can assist in analyzing vast amounts of urban data, generating creative ideas to support urban planners and policymakers for better future planning formulation. On the other hand, UFM can also enhance the participatory planning process [152] by mining public feedback from various digital platforms, making urban planning more inclusive and responsive to the community's needs.

- **Energy management.** The application of UFM in the domain of energy management holds great promise. As one of the initial

efforts in this area, Huang et al. [44] explore a set of capabilities of existing foundation models (i.e., GPT-4 and GPT-4 Vision), showcasing promising results in tackling operational challenges of power systems. Since urban energy data usually possess high sensitivity and personal privacy information, privacy-preserving methods like federated learning [128] could become an effective solution to fully unleash the potential of foundation models in energy management.

- **Environmental monitoring.** We anticipate that UFM can provide detailed simulations and predictive analytics for urban environments [82, 87, 107]. Recently, Saeid [107] introduced ChatClimate, a retrieval-augmented framework to enhance the professional skills of LLMs in the field of climate analysis by incorporating real-time environmental information. Despite these advancements, the utilization of UFM in broader environmental applications, such as air pollution analysis [39], still remains under-explored.

- **Public safety and security.** UFM have exhibited significant potential to enhance public safety and security in various aspects, e.g., serving as transportation safety experts [148], recognizing urban ground collapse events [40], and offering legal services [18], which provide valuable guidance for citizens and government authorities. However, the application of UFM in broader urban public safety challenges, such as crime analysis and health resource allocation, remains largely untapped, offering promising opportunities to fully explore their potential in this field.

6 CONCLUSION AND FUTURE WORK

In conclusion, this paper has offered a comprehensive review of Urban Foundation Models (UFMs), underscoring their transformative potential in deepening our understanding of urban dynamics and significantly enhancing the realization of Urban General Intelligence (UGI). We begin by defining UFM and identifying the major challenges in their development. Furthermore, our taxonomy of current UFM studies, based on urban data modalities and types, provides a clear framework for ongoing research. Additionally, we discuss the potential applications and beneficial impacts of UFM across various critical urban domains.

Looking ahead, the evolution of UFM promises exciting advancements. We anticipate further research focusing on the integration and analysis of multi-source, multi-granularity and multimodal urban data, supporting an expansive range of urban applications. The development of UFM capable of real-time data processing and providing timely urban insights remains a crucial area of focus. Enhancing their spatiotemporal reasoning capabilities will further amplify their effectiveness in dynamic urban environments. Moreover, as UFM evolve, balancing the utility of urban data with privacy and security considerations will be paramount for the ethical advancement of UGI. We are enthusiastic about future developments in this field and myriad ways in which UFM can contribute to smarter, more resilient, and adaptive urban living.

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